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Journal of Complex Networks - Decision on Manuscript ID CONNET-2026-191

1 message

Journal of Complex Networks <onbehalfof@manuscriptcentral.com> Tue, Sep 1, 2026 at 2:42 PM
Reply-To: matjaz.perc@gmail.com
To: akshaysriram.b@gmail.com

01-Sep-2026

Dear Mr. Bhaskara,

Manuscript ID CONNET-2026-191 entitled "Structural Determinants of Network Robustness Under Probabilistic Attack: A Cross-Topology Analysis Using Quasi-Monte Carlo Estimation" which you submitted to the Journal of Complex Networks, has been reviewed. The comments of the reviewer(s) are included at the bottom of this letter.

The reviewers' comments indicate that substantial revisions are required before the manuscript can be reconsidered for publication in the Journal of Complex Networks. We therefore invite you to revise your manuscript and provide a detailed response to all reviewer comments. Please note that we aim to keep the number of review rounds to a minimum. Accordingly, we ask that you address or respond to all points raised by the reviewers, even in cases where you may not fully agree with their comments.

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Because we are trying to facilitate timely publication of manuscripts submitted to the Journal of Complex Networks, your revised manuscript should be uploaded as soon as possible. If it is not possible for you to submit your revision in a reasonable amount of thime, we may have to consider your paper as a new submission.

Once again, thank you for submitting your manuscript to the Journal of Complex Networks and I look forward to receiving your revision.

With best wishes,

REMINDER: if you have not already done so, when submitting your revised manuscript, please include a short alternative-text description under the figure legend for each figure. Alt-text is required to ensure the accessibility of our publications. Guidance is available on the journal's Author Guidelines webpage and at https://academic.oup.com/pages/for-authors/journals/preparing-and-submitting-your-manuscript

With best wishes,
Prof. Matjaz Perc
Editor, Journal of Complex Networks
matjaz.perc@gmail.com

Reviewer(s)' Comments to Author:

Reviewer: 1

Comments to the Author
The manuscript presents a cross-topology analysis of network robustness under probabilistic attack. The authors study how different structural classes respond when attacks do not necessarily succeed, and they use quasi-Monte Carlo estimation to evaluate expected robustness under attack uncertainty. The central result is that network topology remains a stable determinant of vulnerability, even when attack reliability is varied.

I find the manuscript timely, clearly within the scope of the Journal of Complex Networks, and suitable for publication after minor revision. The topic is important, the comparison across network classes is useful, and the probabilistic attack perspective adds practical realism to standard robustness analysis. The paper is also generally well organized and provides informative visualizations. I recommend revision.

1. The Introduction and Discussion should better connect the study to the broader literature on robustness, resilience, and signal propagation in complex networks. The review "Signal propagation in complex networks," Ji et al., Phys. Rep. 1017, 1-96 (2023), would fit well where the authors discuss how damage, fragmentation, and targeted perturbations affect network-mediated processes. The review "Networks behind the morphology and structural design of living systems," Gosak et al., Phys. Life Rev. 41, 1-21 (2022), could be useful in the Discussion, especially when broadening the relevance of topology-dependent robustness to biological and living systems. These references would help position the manuscript more strongly within contemporary complex network science.

2. The main novelty should be stated more sharply. The manuscript is strongest when it presents topology type as an explicit variable in probabilistic robustness analysis. This point should be emphasized consistently in the Abstract, Introduction, and Conclusion. The authors should make clearer how their work differs from previous studies that either assume deterministic attacks or focus on individual real-world networks.

3. The choice of network models and parameter settings should be justified more carefully. The comparison is convincing as a first cross-topology study, but the authors should briefly explain why the selected models and parameter choices are representative. It would also be useful to state whether the conclusions are expected to hold for broader ranges of density, clustering, degree heterogeneity, and community structure.

4. The interpretation of practical implications should be slightly moderated. The manuscript suggests useful design principles for network protection, but the analysis is based on stylized network models and selected attack strategies. The authors should clarify that the results identify robust structural tendencies rather than universal prescriptions for all real infrastructure, biological, or social networks.

5. The presentation would benefit from a light editorial revision. Some passages read somewhat strongly or informally for a research article, and several abbreviations and algorithm names may be difficult for readers who are not already familiar with the earlier probabilistic robustness framework. The authors should consider spelling out the key

ideas more often in words and reducing reliance on dense abbreviation-heavy phrasing.

In summary, this is a useful and publishable contribution to the study of network robustness. The manuscript addresses a relevant problem, extends robustness analysis to a more realistic probabilistic attack setting, and shows that structural network class plays a persistent role in vulnerability. I support publication in the Journal of Complex Networks after the authors address the points above.

Reviewer: 2

Comments to the Author
This manuscript studies network robustness under probabilistic attack using the existing RASR framework, PRQMC estimation, and the HBnnsAGP attack strategy. Its central claim is that the fragility ordering `Scale-Free < Real-World < Small-World < Random` remains stable across attack success rate (ASR) conditions. The topic is relevant, and the paper addresses a meaningful question: most previous work focuses either on deterministic attack models or on fixed network instances, whereas this manuscript attempts to treat topology class itself as the experimental variable. The paper is generally well structured, and the main narrative is easy to follow.

My main reservation, however, is that the strongest conclusions are not yet supported by sufficiently strong evidence. At present, the paper demonstrates a stable pattern under the selected network instances, parameter settings, and experimental conditions, but it does not yet establish the broader claim that topology type is an independent and stable determinant of RASR. The following issues make the current version difficult to accept in its present form.

Major comments

1. The main claim is broader than what the current evidence supports.
The manuscript presents topology type as an independent determinant of RASR, but the evidence currently shown is closer to a comparison of a few representative instances than to a robust topology-class-level result. A concrete example is Table 7, where the ordering is reported as invariant across ASR settings. However, this conclusion is effectively based on one BA graph, one WS graph, one ER graph, and one Karate network. For stochastic graph models such as BA, WS, and ER, a single realization per model is not sufficient to establish a general ordering for the whole topology family.

2. The real-world validation is too limited for the level of generalization used in the paper.
The “real-world social” category is represented essentially by the Karate Club network. While this is a classic and useful benchmark, it is a very small and structurally specific example. It can support the statement that the observed ordering is consistent with one canonical real-world network, but it cannot, by itself, justify broader claims about real-world social networks as a category.

3. The stochastic graph models require repeated realizations and statistical support.
Because BA, WS, and ER are random graph models, conclusions drawn from them should be supported by multiple independent realizations under the same parameter settings, together with averages, variability, and some form of statistical evidence. At present, the paper compares multiple ASR settings, attack strategies, and network sizes, but it does not show whether the reported ordering is robust across realizations of the same model. For example, the claim that “the vulnerability ordering holds exactly at every ASR level from 100% to 50% and under random ASR” is potentially interesting, but without repeated realizations it is difficult to know whether this is a stable finding or simply a property of the particular sampled graphs.

4. The manuscript does not sufficiently disentangle topology effects from parameterization effects.
The BA, WS, and ER networks are studied under fixed parameter choices, such as `m=2` for BA, `k=4, p=0.3` for WS, and an ER edge probability chosen to match the Karate network density. These choices matter, because they directly shape degree heterogeneity, clustering, and connectivity patterns, which may in turn drive the reported robustness ordering. For instance, the discussion attributes the findings repeatedly to hub-dominated structure in scale-free networks, but it remains unclear whether the same ordering would hold if the BA attachment parameter, the WS rewiring probability, or the ER density were varied more systematically.

5. The large-scale discussion seems to go beyond what is directly demonstrated by the main probabilistic framework.
The manuscript emphasizes that the reported ordering persists from `N = 34` up to `N = 10,670`. However, from the presentation of the paper, the larger-scale results appear to rely mainly on HDA and ANC analyses, rather than on the full probabilistic-attack RASR/HBnnsAGP framework that is positioned as the core contribution. In other words, the

small-scale results directly support the probabilistic claim, whereas the large-scale section provides more indirect or auxiliary evidence. Statements such as “This persists from N = 34 to N = 10,670” should therefore be phrased more carefully.

Minor comments

1. The wording in the abstract, discussion, and conclusion should be moderated. Phrases such as “topology type as a stable, independent determinant of RASR” are currently stronger than what the evidence establishes.
2. The distinction between results derived from `RASR + HBnnsAGP` and those derived from `ANC/HDA` should be made clearer throughout the paper. At present, these two analytical lines are sometimes discussed too closely together.
3. The random ASR scenario should be described in more detail. For example, it would help to clarify how node-level ASR values are sampled, whether they are fixed per run, and whether different heterogeneous assignments were repeated.
4. The manuscript would benefit from a more concrete discussion of possible application settings, such as infrastructure protection, communication-network vulnerability assessment, or diffusion-control problems.
5. The reference coverage appears somewhat limited, and the engagement with more recent literature could be strengthened. The current bibliography is relatively short, and aside from a few recent reviews or closely related papers, the citation base relies heavily on classic or foundational references. Given that the manuscript aims to position its contribution in probabilistic attack modeling, network robustness, and cross-topology comparison, the authors should broaden the literature review, particularly by incorporating more recent work on complex-network robustness, probabilistic failure or attack models, and critical-node identification. This would help clarify the manuscript’s incremental contribution relative to the current state of the field and would also strengthen the Related Work section.

Recommendation
Overall, the paper addresses a worthwhile problem and contains potentially interesting empirical observations. However, the present version does not yet provide sufficiently strong support for its broadest claims. The manuscript would be substantially strengthened by repeated experiments across multiple realizations, systematic parameter sensitivity analyses, broader real-world validation, and a more restrained interpretation of the results. My recommendation is ”Major Revision“.